Mustafa Sameen

Ph.D. Student · SERMoS Lab, University of Florida
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RESEARCH INTERESTS

I build and evaluate LLM agents and multi-agent simulations that model human behavior. My work spans household decision-making, individual mobility from GPS records, and travel before, during, and after disasters.

EDUCATION

University of Florida
Ph.D. in Civil and Coastal Engineering — Aug 2025 - Present
Advisor: Xilei Zhao
M.S. in Artificial Intelligence Systems — In progress

Colorado College
B.A. in Computer Science and Mathematics, magna cum laude — May 2025
Study abroad: Aquincum Institute of Technology, Budapest

CONFERENCE AND WORKSHOP PAPERS

Look Both Ways: Reconstructing Two Hidden Months of Post-Earthquake Mobility
Mustafa Sameen, Mrunal Vibhute, Xilei Zhao
HuMob workshop, ACM SIGSPATIAL 2026 · PDF

GAUGE: How Large Language Models Recover Mobility Descriptors from Individual Records
Mustafa Sameen
UMFM workshop, ACM SIGSPATIAL 2026 · PDF

GeoAI-Act: GeoAI-based Activity Inference from GPS Trajectories
Shangkun Jiang, **Mustafa Sameen**, Ally Recalde, Xilei Zhao
GeoAI workshop, ACM SIGSPATIAL 2026 · PDF

Design and Validation of a Grid-Based Home Detection via Stay-Time (GHOST) Software for Mobile Location Data
Alessandra Recalde, **Mustafa Sameen**, Xiaojian Zhang, Xilei Zhao
Transportation Research Board 105th Annual Meeting, 2026 · arXiv:2605.20429

PREPRINTS

PEMAND: Persona-Enriched Multi-Agent Negotiation for Household Decision-Making
Yuran Sun*, **Mustafa Sameen***, Yaotian Zhang, Rongguan Gu, Mrunal Vibhute, Chia-yu Wu, Yuanyuan Lei, Xilei Zhao
2026 · *Equal contribution · arXiv:2604.10475

Synthesizing Attitudes, Predicting Actions (SAPA): Behavioral Theory-Guided LLMs for Ridesourcing Mode Choice Modeling
Mustafa Sameen, Xiaojian Zhang, Xilei Zhao
2025 · arXiv:2509.18181

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RESEARCH EXPERIENCE

Graduate Research Assistant Jul 2025 - Present
University of Florida, SERMoS Lab
Build LLM agents for household negotiation and benchmarks that test what models recover from individual mobility records. Use GPS traces to study everyday mobility and how travel changes during evacuations and recovery.

Summer Research Intern Jun - Jul 2025
MIT JTL-Transit Lab
Designed a spatial retrieval pipeline that combines policy documents with GTFS and GIS data for the Chicago Transit Authority. Prototyped LLM agents for reasoning about transit delays and rerouting. Advised by Haris Koutsopoulos and Muhammad Usama.

Undergraduate Researcher Aug 2024 - May 2025
Colorado College, Department of Mathematics and Computer Science
Developed FORMAL, a retrieval-augmented system that translates mathematical statements into Lean 4 and revises its output using feedback from the theorem prover. Advised by Robert Rennie.

Research Fellow May 2023 - May 2025
State of the Rockies, Colorado College
Built NLP workflows for Indigenous oral histories and a GIS storytelling platform that linked themes from those histories with light pollution data at Bears Ears National Monument.

RESEARCH SOFTWARE AND THESIS

GAUGE · Benchmark and code for testing whether LLMs can recover mobility descriptors from individual location records.

GHOST · Open-source Python package for inferring home locations from GPS data using grid-based stay time.

FORMAL · Mathematics thesis on Lean 4 formalization through retrieval and agentic feedback, Colorado College, 2025.

HONORS

Phi Beta Kappa · Euclid Scholar · Thomas Post Rawles Prize, lower division (2022)

ADDITIONAL EXPERIENCE

Software Engineering Intern, SocialTechLab Jun - Aug 2024
Developed SwiftUI features and contributed to AWS image processing and Go services.

Data Analyst, Quad Innovation Alliance Aug - Dec 2022
Analyzed K-12 education data in Python and SQL and presented findings to stakeholders.